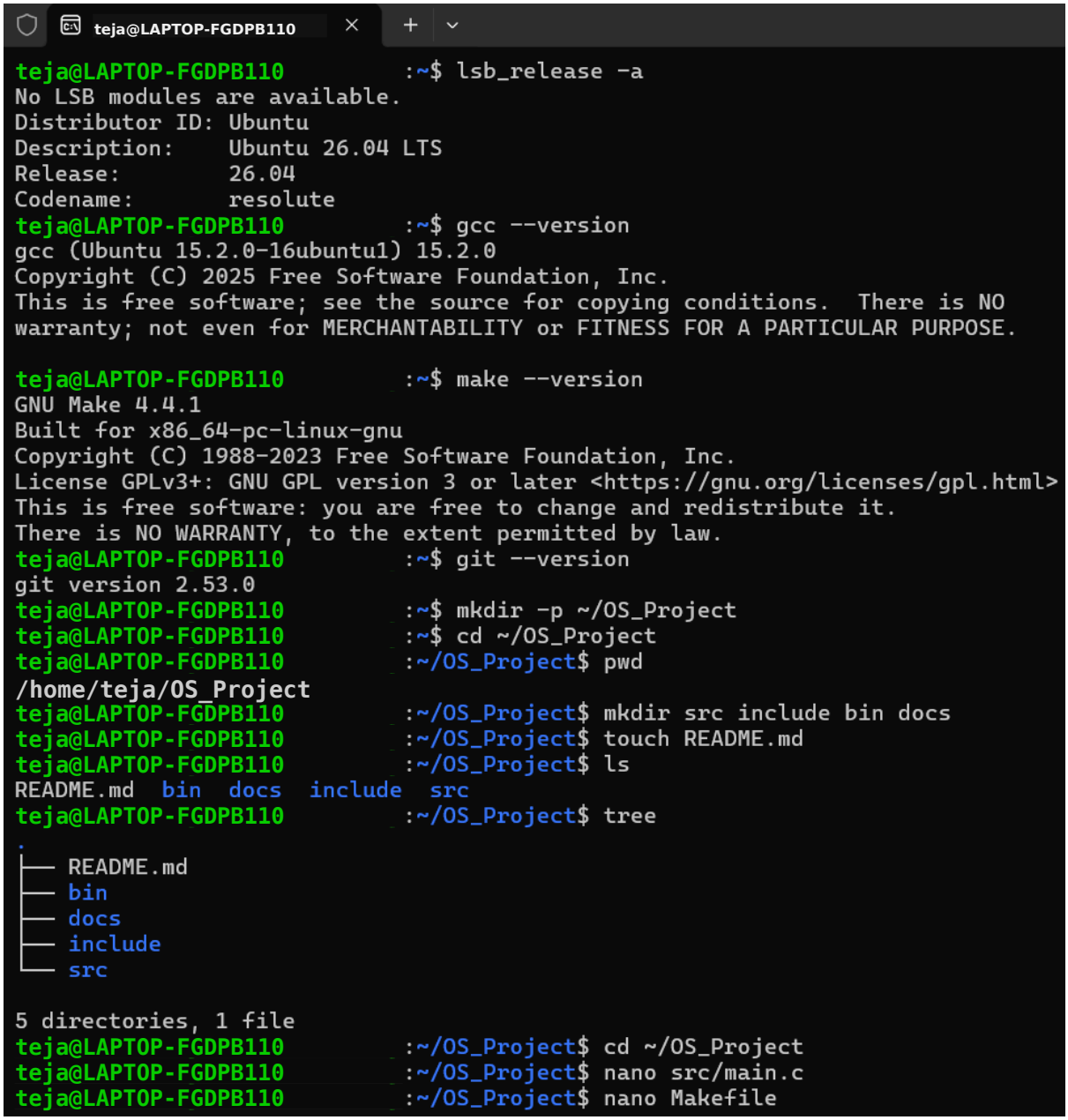
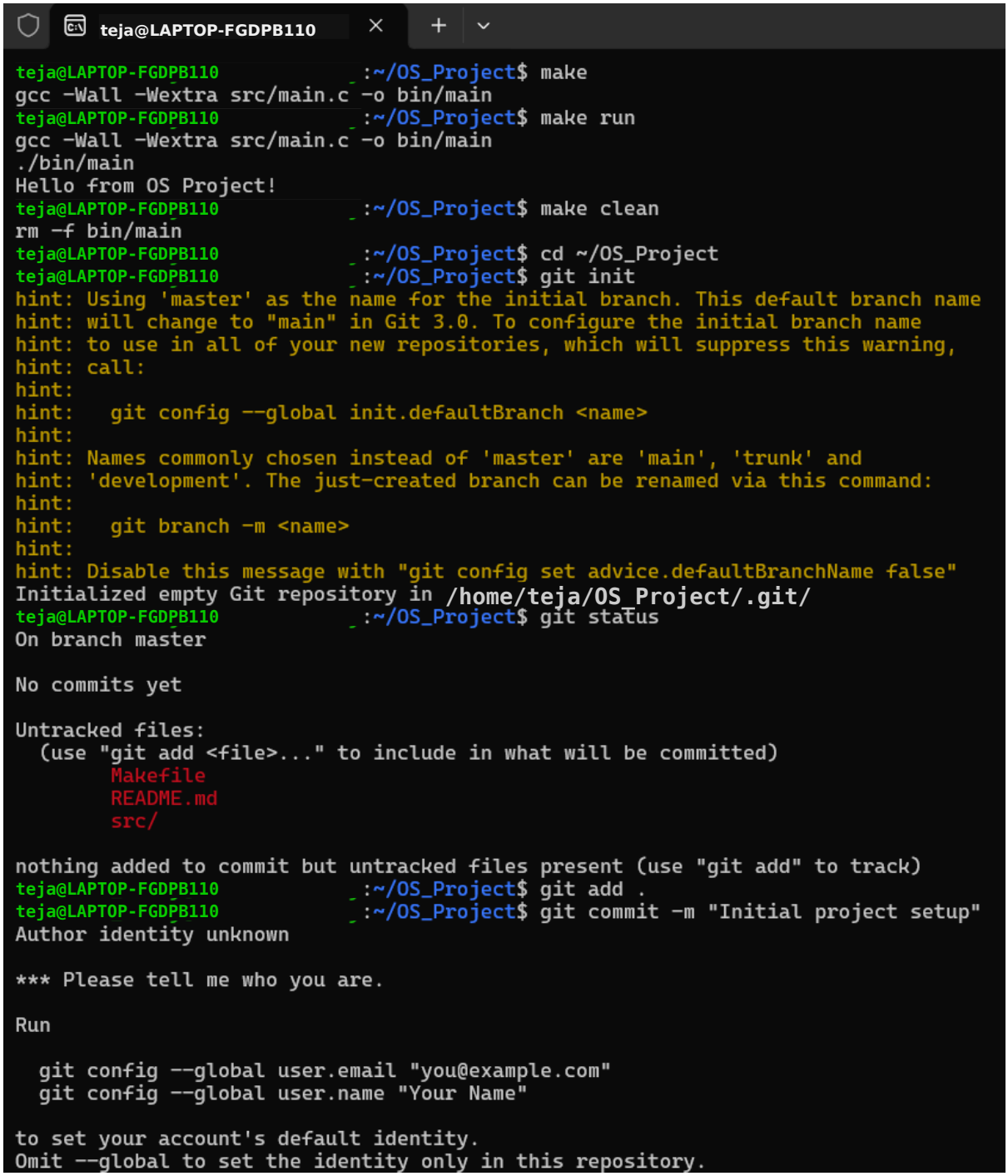
Skill Week-1

Task 1: To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile.




teja@LAPTOP-FGDPB110
teja@LAPTOP-FGDPB110:~/OS_Project$ make
gcc -Wall -Wextra src/main.c -o bin/main
teja@LAPTOP-FGDPB110:~/OS_Project$ make run
gcc -Wall -Wextra src/main.c -o bin/main
./bin/main
Hello from OS Project!
teja@LAPTOP-FGDPB110:~/OS_Project$ make clean
rm -f bin/main
teja@LAPTOP-FGDPB110:~/OS_Project$ cd ~/OS_Project
teja@LAPTOP-FGDPB110:~/OS_Project$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint: git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint: git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/teja/OS_Project/.git/
teja@LAPTOP-FGDPB110:~/OS_Project$ git status
On branch master
No commits yet
Untracked files:
(use "git add <file>..." to include in what will be committed)
Makefile
README.md
src/
nothing added to commit but untracked files present (use "git add" to track)
teja@LAPTOP-FGDPB110:~/OS_Project$ git add .
teja@LAPTOP-FGDPB110:~/OS_Project$ git commit -m "Initial project setup"
Author identity unknown
*** Please tell me who you are.
Run
git config --global user.email "you@example.com"
git config --global user.name "Your Name"
to set your account's default identity.
Omit --global to set the identity only in this repository.

```
nothing added to commit but untracked files present (use "git add" to track)
teja@LAPTOP-FGDPB110           :~/OS_Project$ git add .
teja@LAPTOP-FGDPB110           :~/OS_Project$ git commit -m "Initial project setup"
Author identity unknown

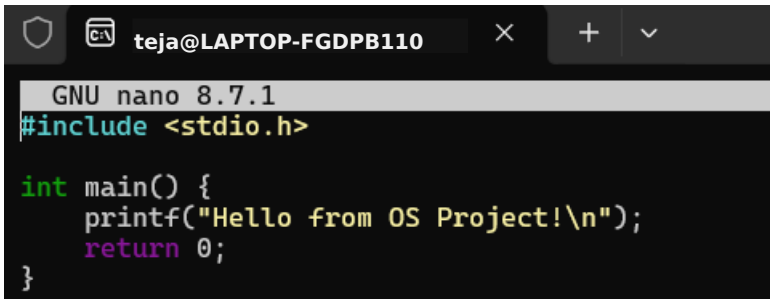
*** Please tell me who you are.

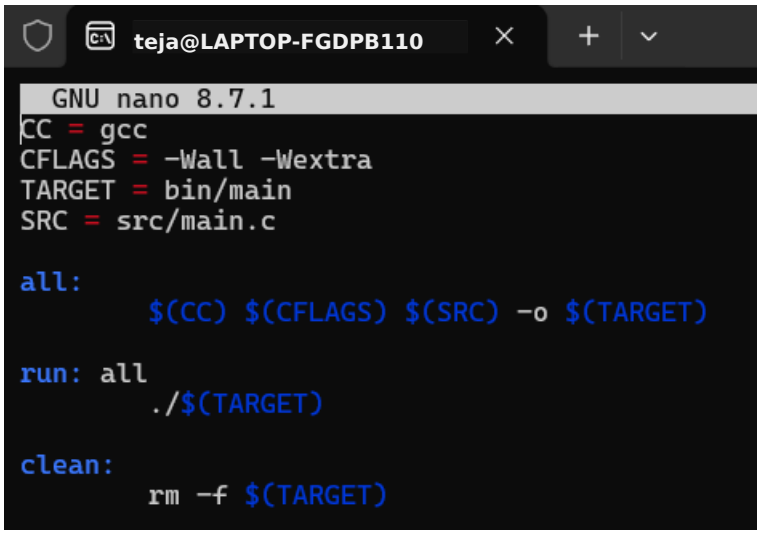
Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

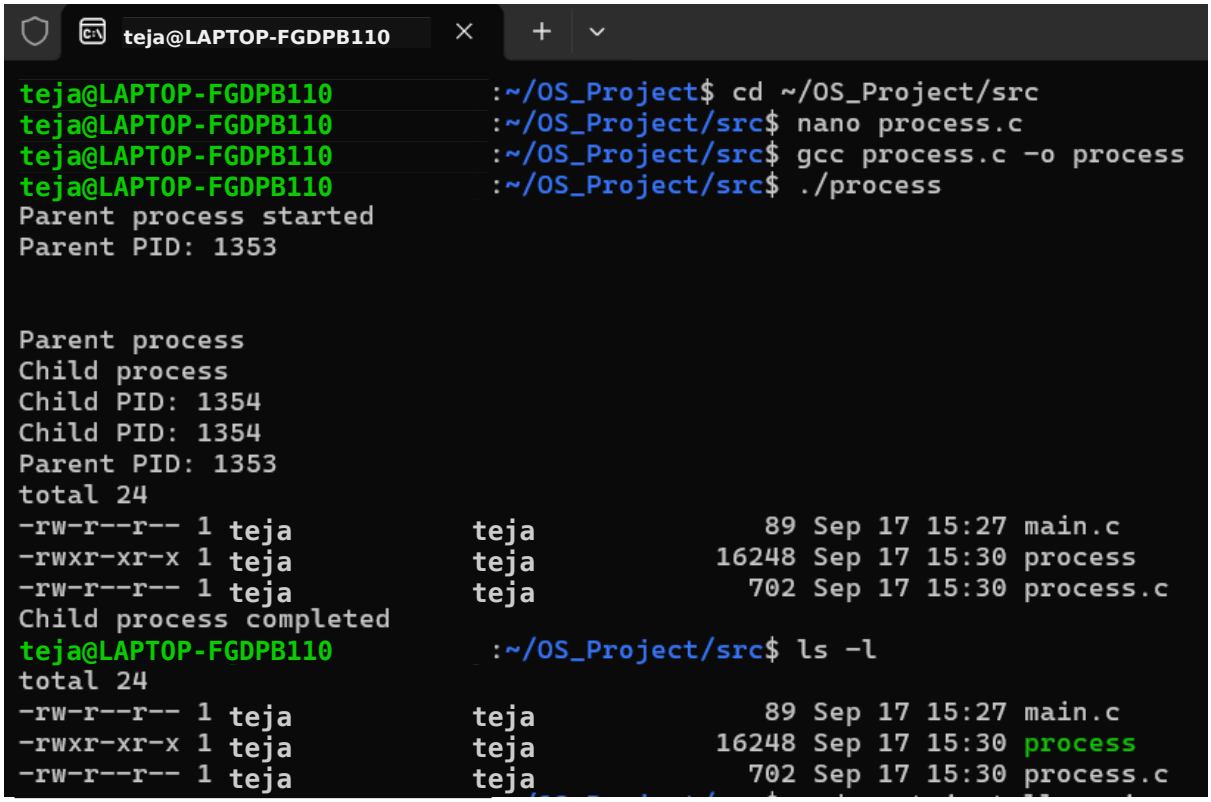
to set your account's default identity.
Omit --global to set the identity only in this repository.

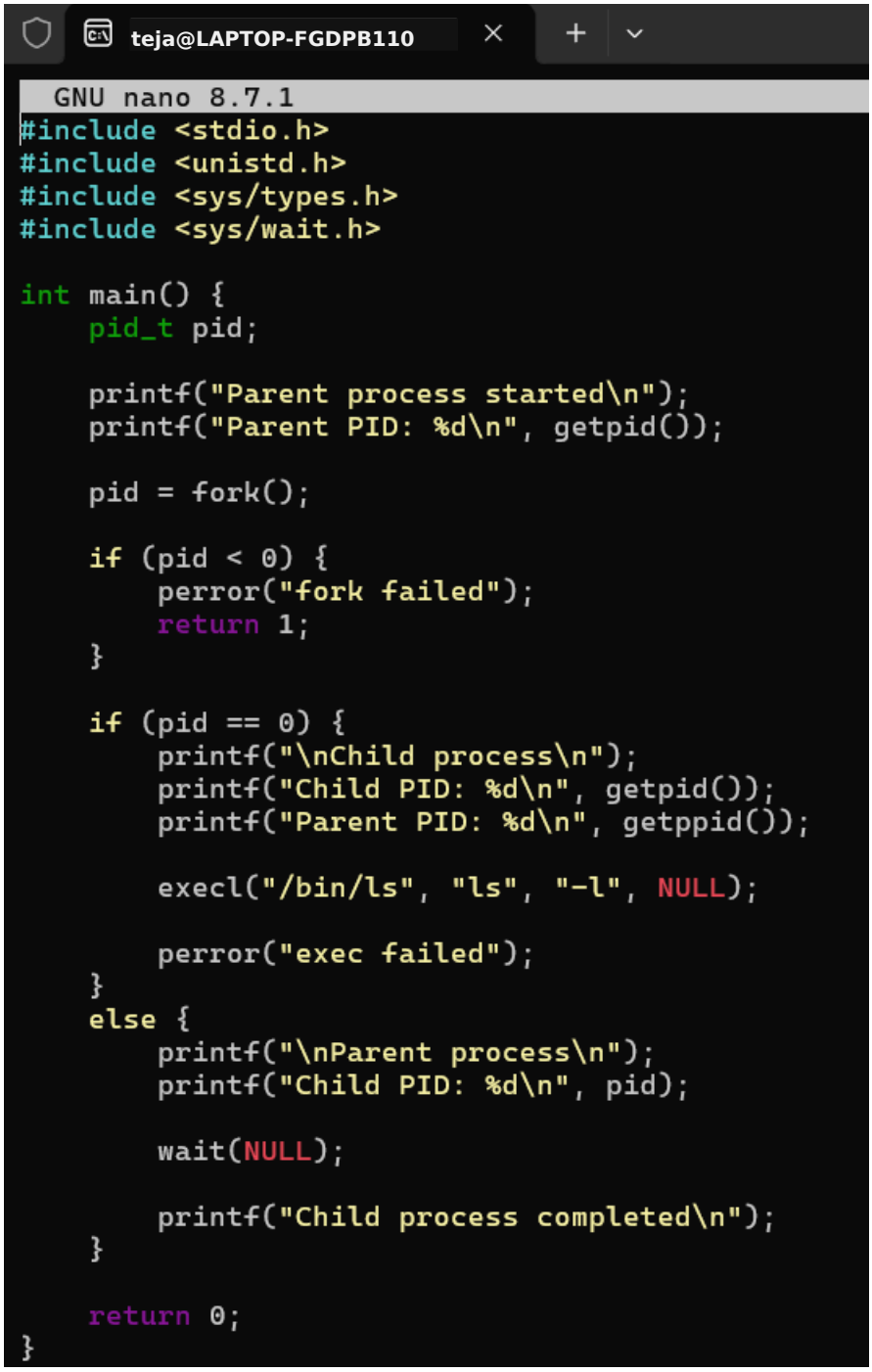
fatal: empty ident name (for <teja@LAPTOP-FGDPB110.localdomain>)     not allowed
teja@LAPTOP-FGDPB110           :~/OS_Project$ git log --oneline
fatal: your current branch 'master' does not have any commits yet
teja@LAPTOP-FGDPB110           :~/OS_Project$
```





Task 2: To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing.


teja@LAPTOP-FGDPB110
teja@LAPTOP-FGDPB110 :~/OS_Project$ cd ~/OS_Project/src
teja@LAPTOP-FGDPB110 :~/OS_Project/src$ nano process.c
teja@LAPTOP-FGDPB110 :~/OS_Project/src$ gcc process.c -o process
teja@LAPTOP-FGDPB110 :~/OS_Project/src$ ./process
Parent process started
Parent PID: 1353
Parent process
Child process
Child PID: 1354
Child PID: 1354
Parent PID: 1353
total 24
-rw-r--r-- 1 teja teja 89 Sep 17 15:27 main.c
-rwxr-xr-x 1 teja teja 16248 Sep 17 15:30 process
-rw-r--r-- 1 teja teja 702 Sep 17 15:30 process.c
Child process completed
teja@LAPTOP-FGDPB110 :~/OS_Project/src$ ls -l
total 24
-rw-r--r-- 1 teja teja 89 Sep 17 15:27 main.c
-rwxr-xr-x 1 teja teja 16248 Sep 17 15:30 process
-rw-r--r-- 1 teja teja 702 Sep 17 15:30 process.c



Task 3: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop.

```
teja@LAPTOP-FGDPB110
teja@LAPTOP-FGDPB110:~/OS_Project/src$ cd ~/OS_Project/src
teja@LAPTOP-FGDPB110:~/OS_Project/src$ nano myshell.c
teja@LAPTOP-FGDPB110:~/OS_Project/src$ gcc myshell.c -o myshell
teja@LAPTOP-FGDPB110:~/OS_Project/src$ ./myshell
myshell> hello
You entered: hello
myshell> Linux
You entered: Linux
myshell> exit
Exiting shell...
teja@LAPTOP-FGDPB110:~/OS_Project/src$
```

```
teja@LAPTOP-FGDPB110
  GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX_INPUT 1024

int main() {
    char input[MAX_INPUT];

    while (1) {
        printf("myshell> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL) {
            printf("\n");
            break;
        }

        input[strcspn(input, "\n")] = '\0';

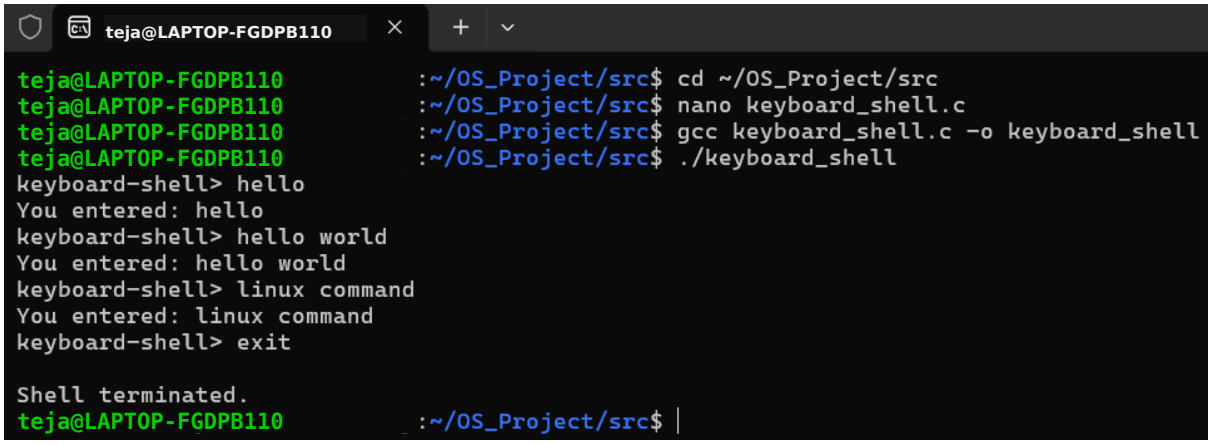
        if (strcmp(input, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }

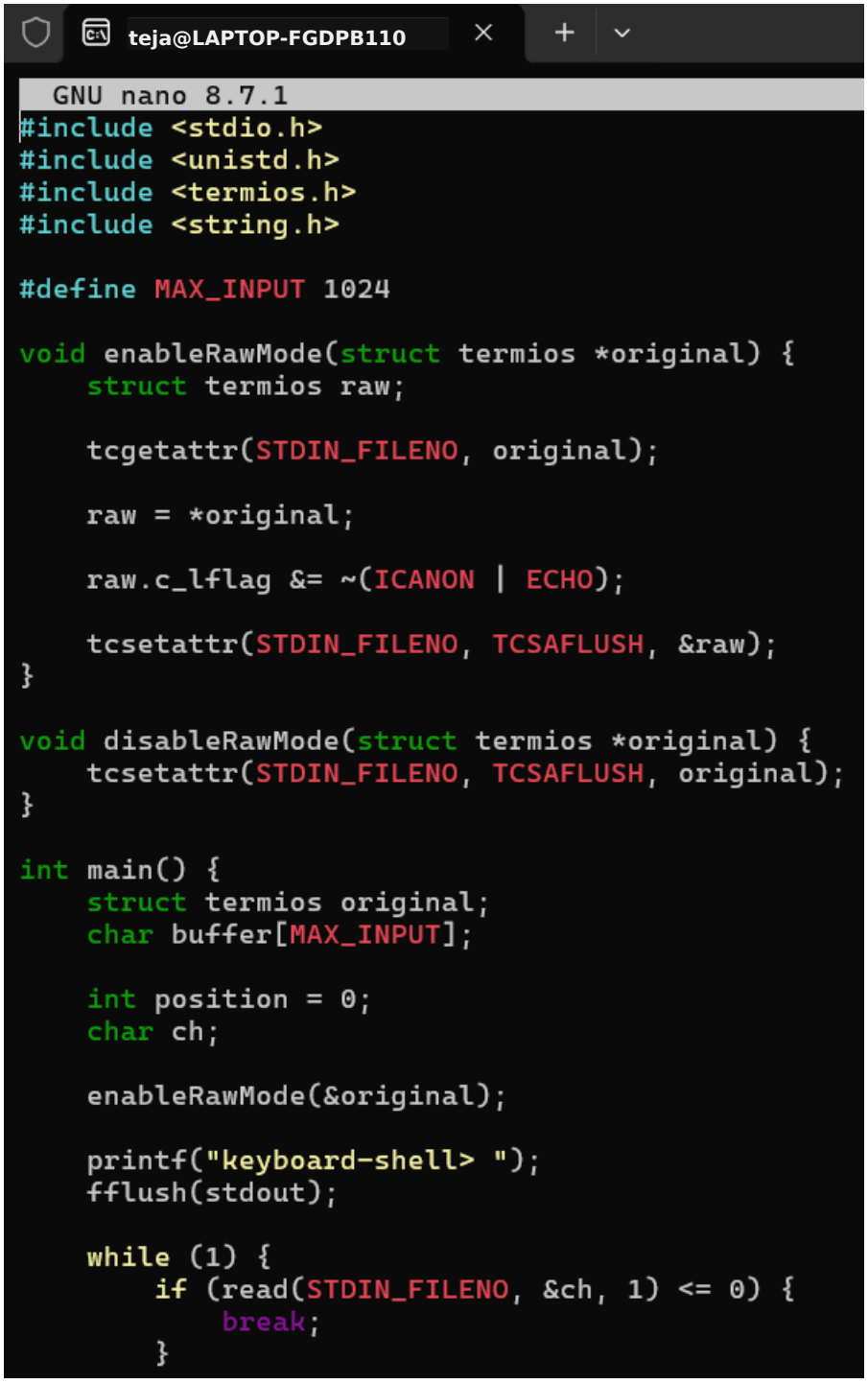
        if (strlen(input) == 0) {
            continue;
        }

        printf("You entered: %s\n", input);
    }

    return 0;
}
```

Task 4: To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction.




teja@LAPTOP-FGDPB110
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <termios.h>
#include <string.h>
#define MAX_INPUT 1024
void enableRawMode(struct termios *original) {
struct termios raw;
tcgetattr(STDIN_FILENO, original);
raw = *original;
raw.c_lflag &= ~(ICANON | ECHO);
tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw);
}
void disableRawMode(struct termios *original) {
tcsetattr(STDIN_FILENO, TCSAFLUSH, original);
}
int main() {
struct termios original;
char buffer[MAX_INPUT];
int position = 0;
char ch;
enableRawMode(&original);
printf("keyboard-shell> ");
fflush(stdout);
while (1) {
if (read(STDIN_FILENO, &ch, 1) <= 0) {
break;
}

```
        /* Enter key */
        if (ch == '\n' || ch == '\r') {
            buffer[position] = '\0';

            printf("\n");

            if (strcmp(buffer, "exit") == 0) {
                break;
            }

            printf("You entered: %s\n", buffer);

            position = 0;
            memset(buffer, 0, sizeof(buffer));

            printf("keyboard-shell> ");
            fflush(stdout);
        }

        /* Backspace */
        else if (ch == 127 || ch == '\b') {
            if (position > 0) {
                position--;

                printf("\b \b");
                fflush(stdout);
            }
        }

        /* Normal character */
        else if (position < MAX_INPUT - 1) {
            buffer[position++] = ch;

            putchar(ch);
            fflush(stdout);
        }
    }

    disableRawMode(&original);

    printf("\nShell terminated.\n");

    return 0;
}
```